

STUDENT GUIDE

How the Internet & the Web Work

Your personal notes — concise, clear, and built for engineers.

Get Techy With Lucky | Instructor: Lucky Nakola

WHAT YOU WILL UNDERSTAND BY THE END

Module 1 — The physical internet: cables, packets, routers, and ISPs

Module 2 — IP addresses, DNS, domain names, and how the web finds any site

Module 3 — HTTP/HTTPS: how your browser talks to a server

Module 4 — How browsers turn code into the pages you see

You are not just a user anymore. After this, you think like an engineer.

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MODULE 1

The Physical Internet

Cables, Packets, Routers & ISPs

The Big Idea

The internet is not magic. It is a physical system — cables, computers, and agreed-upon rules — that moves data around the world as small chunks called packets.

The Analogy — Think of a Post Office

READ THIS FIRST

You write a letter to a friend in Tokyo.

The post office cuts it into 10 smaller pieces, numbers them 1–10, and puts the same destination address on each piece.

Each piece travels a **DIFFERENT** route — some by truck, some by plane.

Your friend's local office reassembles all 10 pieces in order.

Your friend reads the original letter.

That is **EXACTLY** how the internet works.

Your letter = data. The pieces = packets. The post offices = routers.

The Physical Infrastructure

Medium	What It Does
Copper cables	Original wiring. Still used in homes. Carries electrical signals.
Fibre-optic cables	Carries data as pulses of LIGHT through glass. Extremely fast. The internet's main highway.
Submarine cables	Fibre-optic cables on the ocean floor connecting continents. They carry 95%+ of all international traffic.
Wi-Fi / 4G / 5G	Wireless — gets you to the nearest tower or router. The tower connects to fibre cables.
Satellites	Used where cables cannot reach (e.g. remote areas). Higher delay (signal travels to space and back).

ISP — Your On-Ramp to the Internet

An ISP (Internet Service Provider) is the company that sells you internet access. Safaricom, Zuku, and Airtel are Kenyan ISPs. They own the physical cables and towers that connect your device to the global internet.

Packets — The Core Idea

Every file you send or receive — a webpage, a photo, a video — is broken into packets before it travels. Each packet carries:

- The source IP address (where it came from)
- The destination IP address (where it is going)
- A sequence number (so the destination can reassemble in order)
- A payload (the actual piece of data)
- A checksum (to detect errors in transit)

HEADER	PAYLOAD (Data)
Source IP: 41.90.64.1	[Part of a webpage, image, video, email, etc.] (Up to ~1,500 bytes)
Dest IP: 142.250.74.78	
Packet: 0047 of 0312	
Protocol: TCP	
Checksum: A3F7	

TCP vs UDP — Two Ways to Send

Protocol	What It Does & When to Use
TCP	Reliable. Ensures ALL packets arrive and are reassembled in order. If one is lost, it is re-sent. Used for: web pages, emails, file downloads.
UDP	Fast but unreliable. No guarantee of delivery or order. Used for: video calls, live streaming, gaming — where speed matters more than perfection.

Packet Switching — Why the Internet Is Resilient

The internet routes each packet independently through the best available path at that moment. There is no dedicated private road reserved for your data. Every device on the network shares the same

infrastructure — and if one path fails, packets automatically re-route around it. This is why the internet does not have one single point of failure.

KEY TERMS — Module 1

Internet — A global network of networks. Not one company. Not one cable.

ISP — The company that connects you to the internet.

Packet — A small, numbered chunk of data with a header and payload.

TCP — Reliable delivery protocol. All packets, in order.

UDP — Fast delivery protocol. No guaranteed delivery.

Packet Switching — Routing each packet independently. Resilient and efficient.

Latency — The delay between sending and receiving. Measured in milliseconds.

Bandwidth — Maximum data throughput per second (not the same as speed).

MODULE 2

The Digital Address Book

IP Addresses, Routers, DNS & Domain Names

The Big Idea

Computers find each other using numbers (IP addresses), not names. Humans remember names (like google.com), not numbers. DNS is the system that translates one into the other — automatically, in milliseconds, every time you visit a website.

The Analogy — A Phone Book

READ THIS FIRST

Before smartphones, finding someone's phone number meant looking them up in a phone book.
Name on the left — number on the right.
'Nakola, Lucky — 0712 345 678'

DNS works exactly the same way:
Domain name on the left — IP address on the right.
'google.com — 142.250.74.78'

Every time you type a URL, an invisible phone book lookup happens in ~50 milliseconds.

IP Addresses

An IP (Internet Protocol) address is the unique numerical identifier of any device on a network. Every server, phone, laptop, and router has one.

Type	What It Is
IPv4	32-bit. Written as 4 numbers (0–255) separated by dots. e.g. 192.168.1.1 or 142.250.74.78. ~4.3 billion possible addresses. Nearly exhausted.
IPv6	128-bit. Written in hexadecimal groups. e.g. 2001:0db8:85a3::8a2e:0370:7334. Effectively unlimited addresses.
Public IP	Your address on the global internet, assigned by your ISP. Visible to every server you contact.

Type	What It Is
Private IP	Your address on your local network (home Wi-Fi). e.g. 192.168.x.x. Assigned by your router. Not visible on the internet.
Static IP	Fixed. Never changes. Used by servers that must always be at the same address.
Dynamic IP	Changes each session. Automatically assigned by DHCP. Most home users have this.

Routers — The Traffic Directors

A router reads the destination IP address on each packet and decides which path to send it on — like a traffic officer at a junction. Your home router connects your devices to your ISP. Backbone routers in data centres handle billions of packets per second. Routers maintain routing tables — live databases of the best paths to every known network.

DNS — How Names Become Numbers

The DNS Hierarchy

Server	Its Role
Root Servers	Top of the chain. 13 sets worldwide. Know where to find every TLD server.
TLD Servers	Handle top-level domains (.com, .ke, .org). The .ke TLD is managed by KENIC in Kenya.
Authoritative Server	The final answer. Holds the actual IP for your specific domain. You configure this when you buy a domain.
Recursive Resolver	The server your device asks first (usually your ISP's, or Google 8.8.8.8 / Cloudflare 1.1.1.1). It hunts through the chain on your behalf and caches results.

The DNS Resolution — Step by Step

```
You type: www.gettechy.co.ke — and press Enter.
```

1. Browser checks its own cache — has it looked this up recently?
2. OS checks its DNS cache and the local 'hosts' file.
3. Device asks the Recursive Resolver: 'What is the IP for gettechy.co.ke?'
4. Resolver asks a Root Server: 'Who handles .ke domains?'
Root replies: 'KENIC does. Here is their address.'
5. Resolver asks KENIC (.ke TLD): 'Who handles gettechy.co.ke?'
KENIC replies: 'The authoritative server is ns1.host.co.ke'


```
6. Resolver asks ns1.host.co.ke:      'What is the IP for
www.gettechy.co.ke?'
   Auth server replies:                '196.201.214.50'
7. Resolver caches the result (for the TTL duration) and returns the IP.
8. Browser now has the IP. It can send the HTTP request.
```

Total time: approximately 20-120 milliseconds.

Domain Names — The Structure

```
www . gettechy . co . ke
|   |   |   |
|   |   |   └─ Country Code TLD (.ke = Kenya)
|   |   └─── Second-level TLD (.co = commercial)
|   └────── Your unique domain name (gettechy)
└────────── Subdomain (www)
```

DNS Records — The Most Important Ones

Record Type	What It Does
A Record	Maps a domain to an IPv4 address. gettechy.co.ke → 196.201.214.50
AAAA Record	Maps a domain to an IPv6 address.
CNAME	An alias. www.gettechy.co.ke → gettechy.co.ke (points one name to another name)
MX Record	Points to the mail server. Controls where email for your domain is delivered.
TXT Record	Text data. Used for domain verification, Google, SSL checks, spam prevention.
TTL	Time to Live — how long resolvers cache your record before checking again. Low TTL = faster propagation of changes.

KEY TERMS — Module 2

IP Address — The unique number identifying a device on a network.

DNS — Domain Name System. Translates domain names into IP addresses.

Domain Name — The human-friendly web address (e.g. gettechy.co.ke).

TLD — Top-Level Domain. The last part of a domain: .com, .ke, .org.

A Record — DNS entry that maps a domain to an IP address.

TTL — How long a DNS result is cached before it expires.

Router — Device that reads IP addresses and forwards packets toward their destination.

DHCP — Protocol that automatically assigns IP addresses on a network.

MODULE 3

The Conversation

The Client-Server Model, HTTP/HTTPS, Requests & Responses

The Big Idea

The web is a conversation between two parties: a client (your browser) and a server (the computer hosting the website). HTTP is the language they use. Every page load is a request-response exchange — your browser asks, the server answers.

The Analogy — A Restaurant

READ THIS FIRST

You = the CLIENT (your browser)

The waiter = HTTP (the agreed language of communication)

The kitchen= the SERVER (stores and prepares the data)

The menu = the URL (what is available and where to find it)

Your order = the HTTP REQUEST

The food = the HTTP RESPONSE

200 OK = 'Here is your food.'

404 Not Found = 'Sorry, we do not have that on the menu.'

500 Server Error = 'The kitchen is on fire.'

Client vs Server

Role	What It Does
Client	Any device or software that sends a REQUEST. Your browser, a mobile app, or a command-line tool. Clients always initiate the conversation.
Server	A computer that listens for requests, processes them, and sends back a RESPONSE. Runs software like Apache, Nginx, or Node.js.
Key Rule	Clients ask. Servers answer. A server never sends data unless a client requests it first.

The HTTP Request

When you visit a URL, your browser builds and sends an HTTP request. Here is what it contains:

```
REQUEST LINE:
GET /courses/javascript HTTP/1.1
|           |           |_____ Protocol version
|           |_____ Resource path being requested
|_____ HTTP Method (the action)

HEADERS (metadata):
Host: www.gettechy.co.ke
Accept: text/html
User-Agent: Chrome/125
Cookie: session_id=abc123

BODY (only for POST, PUT, PATCH):
{ 'name': 'Lucky', 'email': 'lucky@example.com' }
```

HTTP Methods

Method	Purpose & When to Use
GET	Retrieve data. No body. Use for: loading pages, fetching API data.
POST	Send new data. Has a body. Use for: login forms, registrations, creating records.
PUT	Replace an entire resource with new data. Use for: full profile updates.
PATCH	Partially update a resource. Use for: changing just one field.
DELETE	Remove a resource. Use for: deleting accounts or records.

The HTTP Response

```
STATUS LINE:
HTTP/1.1 200 OK
|           |           |_____ Human-readable message
|           |_____ Status code
|_____ Protocol version

HEADERS:
Content-Type: text/html; charset=UTF-8
Content-Length: 48291
Set-Cookie: session_id=xyz789; HttpOnly; Secure
Cache-Control: max-age=3600

BODY:
<!DOCTYPE html>
<html><body><h1>Welcome!</h1></body></html>
```

HTTP Status Codes — Memorise These

Group	Most Important Code	Others to Know
2xx — Success	200 OK — request succeeded, here is the data.	201 Created, 204 No Content
3xx — Redirect	301 Moved Permanently — URL has changed forever.	302 Found (temp), 304 Not Modified
4xx — Client Error	404 Not Found — resource does not exist.	401 Unauthorised, 403 Forbidden, 429 Too Many Requests
5xx — Server Error	500 Internal Server Error — server broke.	502 Bad Gateway, 503 Service Unavailable

401 vs 403 — Know the Difference

401 Unauthorised — 'Who are you? You have not logged in yet.'

403 Forbidden — 'I know who you are — but you are not allowed here.'

401 = Authentication failed (not logged in).

403 = Authorisation failed (logged in, but wrong permissions).

HTTPS — Adding Encryption

HTTP sends everything as plain text. Anyone monitoring the network can read your password, your data, everything. HTTPS adds TLS (Transport Layer Security) encryption, which scrambles all data in transit so only the client and server can read it.

Concept	What It Means
TLS/SSL	The encryption protocol under HTTPS. Creates an encrypted tunnel. Even if someone intercepts your packets, they see random noise.
SSL Certificate	A digital ID card issued to a server by a trusted Certificate Authority. Proves the server is who it says it is. Enables HTTPS.
The padlock icon	Shown in your browser address bar. Means the site has a valid certificate and the connection is encrypted.
Mixed content	A page loaded over HTTPS that also loads some resources (images, scripts) over HTTP. Browsers warn or block this — it weakens security.

KEY TERMS — Module 3

HTTP	— HyperText Transfer Protocol. The rules for web communication.
HTTPS	— HTTP with TLS encryption. Required for any secure site.
TLS	— Transport Layer Security. The encryption protocol behind HTTPS.
Request	— What the client sends: method + URL + headers + (optional) body.
Response	— What the server sends back: status code + headers + body.
Status Code	— Three-digit number indicating the outcome of a request.
Header	— Metadata attached to a request or response.
Cookie	— Data the server sends to the browser to store and return on future requests.

MODULE 4

The Canvas

How Browsers Transform HTML, CSS & JavaScript into What You See

The Big Idea

Once your browser receives the server's HTML, CSS, and JavaScript, it has to turn that raw code into a visual, interactive interface. This process is called the Critical Render Path — and understanding it makes you a better developer.

The Analogy — Building a House

READ THIS FIRST

HTML = The architect. Defines the structure: walls, rooms, windows, doors.

CSS = The interior designer. Decides colours, textures, sizes, and layout.

JS = The electrician. Makes things move, respond, and interact.

The browser is the CONSTRUCTION CREW that takes all three and builds the house.

Without HTML — nothing to build on.

Without CSS — functional but ugly.

Without JS — beautiful but completely static.

The Critical Render Path — 6 Steps

This is the sequence of steps the browser must complete before you see ANYTHING on screen.

```
STEP 1 — Parse HTML → Build the DOM
Browser reads HTML from top to bottom.
Converts every tag into a node. Assembles nodes into a tree: the DOM.
```

```
<html>                DOM Tree:
  <body>               Document
    <h1>Hello</h1>    └─ html
    <p>World</p>      └─ body
                     └─ h1 ('Hello')
</body>              └─ p  ('World')
</html>
```

```
STEP 2 — Parse CSS → Build the CSSOM
```

Browser reads all CSS. Builds the CSSOM (CSS Object Model) — same tree structure.

⚠ CSS is RENDER-BLOCKING. The browser will not show anything until all CSS is parsed.

STEP 3 — Merge DOM + CSSOM → Render Tree

Combines both trees. Only VISIBLE elements are included.

Elements with `display:none` are excluded from the Render Tree.

STEP 4 — Layout (Reflow)

Calculates the exact position and size of every element on screen.

Uses the CSS Box Model: content + padding + border + margin.

STEP 5 — Paint

Converts layout information into actual pixels.

Handles text, colours, borders, shadows, images.

STEP 6 — Composite

Assembles painted layers in the correct order.

Your GPU does this step — why CSS animations using `transform/opacity` are smooth.

JavaScript — The Dynamic Layer

JavaScript is executed by the browser's JS engine (V8 in Chrome). By default, a `<script>` tag blocks HTML parsing — the browser stops, downloads the script, runs it, then continues.

Script Type	Behaviour
<code><script></code>	BLOCKS HTML parsing. Downloads and executes immediately. Avoid in <code><head></code> .
<code><script defer></code>	Downloads in parallel. Executes AFTER HTML is parsed. Does NOT block. Best choice for most scripts.
<code><script async></code>	Downloads in parallel. Executes as soon as download finishes — can interrupt parsing. Best for independent scripts (e.g. analytics).

Browser Components — What Is Under the Hood

Component	What It Does
Rendering Engine	Reads HTML and CSS, builds the Render Tree, and paints the screen. Blink (Chrome/Edge), Gecko (Firefox), WebKit (Safari).
JavaScript Engine	Executes JS code. V8 in Chrome (also powers Node.js), SpiderMonkey in Firefox.

Component	What It Does
Network Layer	Handles all HTTP/HTTPS requests — fetching HTML, CSS, JS, images, fonts.
Storage	Manages cookies, localStorage, sessionStorage, and browser cache.

The Full Journey — All Four Modules Connected

```
You type: https://www.gettechy.co.ke/javascript — press Enter.

[MODULE 1 — NETWORK]
Data will travel as TCP/IP packets through physical cables.
Safaricom's fibre → data centres → submarine cables if needed.

[MODULE 2 — DNS]
Browser checks cache → Recursive Resolver → Root → .ke TLD → Auth Server
Auth Server replies: 196.201.214.50    (~50ms)

[MODULE 3 — HTTP]
TLS handshake — encrypted connection established.
Browser sends:  GET /javascript HTTP/1.1
Server replies: HTTP/1.1 200 OK  +  HTML body

[MODULE 4 — BROWSER]
HTML received → DOM built.
CSS received (blocks) → CSSOM built.
DOM + CSSOM → Render Tree → Layout → Paint → Composite.
You see the page.    (~Total: 200-800ms from keypress to pixels)
```

KEY TERMS — Module 4

DOM — Document Object Model. The browser's internal map of the HTML page.

CSSOM — CSS Object Model. The browser's internal map of all CSS rules.

Render Tree — DOM + CSSOM merged. Only visible elements. Used for layout and paint.

Reflow — Recalculating element positions and sizes. Triggered by DOM/CSS changes.

Repaint — Redrawing pixels after a visual change (no position change).

Render-blocking — A resource that prevents the browser from painting the page until it loads.

Critical Render Path — The sequence of steps from HTML bytes to visible pixels.

V8 / Blink — Chrome's JavaScript engine and rendering engine respectively.

Quick-Reference Terminology Vault

Every term below is one you will use constantly as a web developer. Do not just memorise the definitions — understand what problem each one solves.

Infrastructure

Term	Definition
Internet	A global network of networks. Owned by nobody, used by everyone.
ISP	The company that connects you to the internet (Safaricom, Zuku, Airtel).
Packet	A small, numbered data chunk. All internet data travels as packets.
TCP	Reliable delivery. All packets arrive and are reassembled in order.
UDP	Fast delivery. No guarantee of arrival or order. Used for streaming/gaming.
Router	Reads IP addresses and forwards packets toward their destination.
Fibre-optic	Glass cables carrying data as light pulses. The internet's main highway.
Latency	The delay between sending a request and receiving a response (ms).
Bandwidth	Maximum data throughput per second. Not the same as speed.

Addressing & DNS

Term	Definition
IP Address	Unique number identifying a device on a network. e.g. 142.250.74.78
IPv4	32-bit address format. ~4.3 billion addresses. e.g. 192.168.1.1
IPv6	128-bit address format. Effectively unlimited addresses.
DNS	Domain Name System. Translates domain names to IP addresses.
Domain name	Human-readable address. e.g. gettechy.co.ke
TLD	Top-Level Domain — the last segment of a domain (.com, .ke, .org).
A Record	DNS record mapping a domain to an IPv4 address.
CNAME	DNS alias — points one domain name to another domain name.
TTL	Time to Live — how long a DNS result is cached before expiring.
CDN	Content Delivery Network — global servers that cache your site near users.
Hosting	Renting server space where your website files live.

HTTP & Web Architecture

Term	Definition
HTTP	HyperText Transfer Protocol. The language of the web.
HTTPS	HTTP with TLS encryption. Required for secure sites.
TLS/SSL	Encryption protocol under HTTPS. Scrambles data in transit.
SSL Certificate	Digital ID proving a server's identity. Enables HTTPS.
Request	What the client sends: method + URL + headers + body.
Response	What the server sends: status code + headers + body.
Status Code	Three-digit result of an HTTP request. 200 OK, 404 Not Found, 500 Error.
Cookie	Data stored in the browser and sent back to the server on every request.
Session	Server-side storage of user state. Identified by a session ID.
Cache	Stored copy of a resource to avoid repeated downloads.
API	Application Programming Interface. A set of endpoints for software to communicate.
JSON	JavaScript Object Notation. Lightweight text format for data exchange.
DOM	Document Object Model. The browser's live map of the HTML page.
CSSOM	CSS Object Model. The browser's map of all CSS rules.
Render Tree	DOM + CSSOM merged. Only visible elements. Used to calculate layout.
Reflow	Recalculating element positions after DOM or CSS changes.

Self-Assessment Quiz

Answer these questions on your own first. Then review the answers below. Be honest with yourself — the goal is to find what you do not yet understand, not to feel good about what you already do.

Multiple Choice Questions

Circle or write the letter of the best answer.

1. Amira is streaming a video when the undersea cable route to the server is cut. What most likely happens?
 - A) The video stops immediately and cannot resume until the cable is repaired.
 - B) Her router automatically downloads a backup video from a local cache.
 - C) The network re-routes her video packets through an alternative path with minimal interruption.
 - D) The video continues because she already downloaded the full file.
2. A developer updates their A Record on DNS but can still see the old site. Their colleague in Kisumu sees the new site. What is the cause?
 - A) The developer made an error in the record.
 - B) The developer's DNS resolver has a cached copy of the old IP — TTL has not expired yet.
 - C) The developer's browser cached the old HTML.
 - D) The new server is blocking the developer's IP address.
3. A student logs into a school portal and then tries to access /admin/dashboard. The server denies access. Which status code is correct?
 - A) 404 Not Found
 - B) 401 Unauthorised
 - C) 403 Forbidden
 - D) 302 Redirect
4. A webpage takes 6 seconds to show anything on screen. The HTML is tiny but the `<head>` links to 4 large CSS files and 3 JS files without 'defer'. What is the most likely cause?
 - A) The server is too slow.
 - B) The images are too large and blocking the initial render.
 - C) The CSS and JS files are render/parser-blocking — the browser waits for all of them before painting.

- D) The browser ran out of memory.
5. A junior developer says: 'Our login form is safe — we use POST so the password is in the body, not the URL.' What is wrong with this reasoning?
- A) Nothing — POST requests are always encrypted.
 - B) POST hides data from the URL but does NOT encrypt it. Without HTTPS, the body is plain text, readable by anyone monitoring the network.
 - C) GET should be used for login forms, not POST.
 - D) Modern routers encrypt all POST requests automatically.

Answer Key

Answers

- Q1 → C — Packet-switched networks automatically re-route around failed paths. This is by design.
- Q2 → B — DNS TTL caching. The resolver is serving a stale cached record. DNS propagation takes time.
- Q3 → C — 403 Forbidden means authenticated (logged in) but not authorised (no admin permission).
- 401 = not logged in at all. 403 = logged in but no permission. Know the difference.
- Q4 → C — CSS is render-blocking. JS without defer is parser-blocking. Both must fully load before the browser can paint anything.
- Q5 → B — POST only hides data from the URL bar. Without HTTPS, the request body is plain text.
- Anyone on the same Wi-Fi can intercept and read it. HTTPS encrypts EVERYTHING.

Open-Ended Questions

Write your answers out fully. Use the correct terminology. These are the kinds of questions a real interviewer or client will ask you.

Question 1 — Trace a 404 Error

Trace every step that happens when a user types this URL and gets a 404 error:

<https://www.learningportal.co.ke/courses/machine-learning>

Your answer must cover: DNS resolution, TLS handshake, HTTP request, server processing, HTTP response (including the 404 status), and browser rendering of the error page.

Question 2 — Performance Debug

A website loads in 8 seconds on a Kenyan mobile phone. Users are leaving.

Identify at least one bottleneck at EACH layer (network, DNS, HTTP, browser rendering)

and propose a specific fix for each one.

Question 3 — Security Incident

Students at a school notice their grades are being changed. The school portal uses HTTP.

Explain exactly how an attacker on the school Wi-Fi could be doing this.

What must the school fix, and why does it fix the problem?

The Engineer's Debugging Mindset

Whenever something breaks on the web — and it will — trace the layers in order. You now know every layer.

```
STEP 1 — Is it a NETWORK issue?  
Can you ping the server? Is there internet connectivity?  
Tools: ping, traceroute, checking your Wi-Fi connection.  
  
STEP 2 — Is it a DNS issue?  
Is the domain resolving to the right IP?  
Tools: nslookup gettechy.co.ke or dnschecker.org  
  
STEP 3 — Is it an HTTP issue?  
What status code is the server returning? What do the headers say?  
Tools: Browser DevTools → Network tab. Look at every request.  
  
STEP 4 — Is it a RENDERING issue?  
Are there JavaScript errors? CSS not loading? DOM not building?  
Tools: Browser DevTools → Console tab. Look for red errors.  
  
Every problem has a layer. Find the layer. Fix the problem.
```

You now see what 99% of people using the internet never see.
Use it well. Build something great.

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